

7. Deployment & DevOps Checklist

Target Goal: Deploy the frontend PWA, FastAPI AI backend, and managed database to production with SSL, CDN caching, and automated scaling.

Step 1: Frontend Deployment to Vercel or Cloudflare Pages (Free Tier)

1. Push your code to GitHub: `git init && git add . && git commit -m "Initial commit" && git push`
2. Log into [Vercel.com](https://vercel.com) or Cloudflare Pages.
3. Click **New Project** and import your GitHub repository.
4. Build settings:
 - **Framework Preset:** Vite
 - **Build Command:** `npm run build`
 - **Output Directory:** `dist`
5. Click **Deploy**. Your PWA is live with automated global SSL within 60 seconds!

Step 2: Backend Containerization with Docker

Save this `Dockerfile` in your backend directory:

```
# backend/Dockerfile
FROM python:3.11-slim

WORKDIR /app

# Install system dependencies for OpenCV and image processing
RUN apt-get update && apt-get install -y --no-install-recommends \
    libgl1-mesa-glx libglib2.0-0 curl \
    && rm -rf /var/lib/apt/lists/*

COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

COPY . .

# Run as non-root user for security
USER 1000

EXPOSE 8000
CMD ["uvicorn", "main:app", "--host", "0.0.0.0", "--port", "8000"]
```

Step 3: Deploy Backend on Render.com or Railway

1. Log into [Render.com](#) \$
ightarrow\$ **New Web Service**.

2. Connect your backend repository and select **Docker runtime**.

3. Configure environment variables:
 - `DATABASE_URL=postgresql://user:password@host/cropcare`
 - `ALLOWED_ORIGINS=https://your-frontend-domain.com`
 - `MODEL_PATH=/app/yolov8n-crop-diseases.pt`

4. Deploy. You receive an automated HTTPS endpoint: `https://cropcare-api.onrender.com` .

Step 4: Production Pre-Flight Checklist

Task	Status	Validation Check
HTTPS & SSL	Critical	PWA Service Workers & Camera access require active SSL (https://).
CORS Security	Critical	Ensure <code>allow_origins</code> in FastAPI matches your production frontend URL only.
Rate Limiting	High	Install <code>slowapi</code> on FastAPI to limit scans to 30 requests / minute per IP.
Lighthouse PWA Audit	High	Run Chrome Lighthouse. Target ≥ 90 on Performance and PWA criteria.
Error Tracking (Sentry)	Medium	Install <code>@sentry/react</code> to receive automated alerts if farmer scans crash.